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Measurement Descriptions and Units

Real Time files generally contain the last 45 days of "Realtime" data - data that went through automated quality checks and were distributed as soon as they were received. Historical files have gone through post-processing analysis and represent the data sent to the archive centers. The formats for both are generally the same, with the major difference being the treatment of missing data. Missing data in the Realtime files are denoted by "MM" while a variable number of 9's are used to denote missing data in the Historical files, depending on the data type (for example: 999.0 99.0).

General

Units: Station pages display the current hour's measurements in Imperial units by default, but can be changed by the viewer to metric units. When accessing Real Time and Historical data files, the **measurements are generally in metric units**, as described below, and cannot be changed.

Time: Station pages show current observations in station local time by default, but can be changed by the viewer to UTC (formerly GMT). **Both Realtime and Historical files show times in UTC only.** See the [Acquisition Time help topic](#) for a more detailed description of observation times. For more information on the times in the files, see the [changes page](#).

Station ID: Five-digit WMO [Station Identifier](#), used since 1976. ID's can be reassigned to future deployments within the same 1 degree square.

Formats: Data are classified according to the following groups. The header lines are shown at the beginning of group. Note that in the Realtime files, non-data lines begin with "#". Such lines should be treated as comment lines.

Standard Meteorological Data

#YY	MM	DD	hh	mm	WDIR	WSPD	GST	WVHT	DPD	APD	MWD	PRES	ATMP	WTMP	DEW
#yr	mo	dy	hr	mn	degT	m/s	m/s	m	sec	sec	degT	hPa	degC	degC	deg
2014	09	11	16	50	120	5.0	6.0	0.6	6	4.2	134	1016.5	29.3	30.5	24.1



WDIR

Wind direction (the direction the wind is coming from in degrees clockwise from true N) during the same period used for WSPD. See [Wind Averaging Methods](#)

WSPD

Wind speed (m/s) averaged over an eight-minute period for buoys and a two-minute period for land stations. Reported Hourly. See [Wind Averaging Methods](#).

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GST	Peak 5 or 8 second gust speed (m/s) measured during the eight-minute or two-minute period. The 5 or 8 second period can be determined by payload, See the Sensor Reporting, Sampling, and Accuracy section.
WVHT	Significant wave height (meters) is calculated as the average of the highest one-third of all of the wave heights during the 20-minute sampling period. See the Wave Measurements section.
DPD	Dominant wave period (seconds) is the period with the maximum wave energy. See the Wave Measurements section.
APD	Average wave period (seconds) of all waves during the 20-minute period. See the Wave Measurements section.
MWD	The direction from which the waves at the dominant period (DPD) are coming. The units are degrees from true North, increasing clockwise, with North as 0 (zero) degrees and East as 90 degrees. See the Wave Measurements section.
PRES	Sea level pressure (hPa). For C-MAN sites and Great Lakes buoys, the recorded pressure is reduced to sea level using the method described in <i>NWS Technical Procedures Bulletin 291</i> (11/14/80). (labeled BAR in Historical files)
ATMP	Air temperature (Celsius). For sensor heights on buoys, see Hull Descriptions . For sensor heights at C-MAN stations, see C-MAN Sensor Locations
WTMP	Sea surface temperature (Celsius). For buoys the depth is referenced to the hull's waterline. For fixed platforms it varies with tide, but is referenced to, or near Mean Lower Low Water (MLLW) .
DEWP	Dew point temperature taken at the same height as the air temperature measurement.
VIS	Station visibility (nautical miles). Note that buoy stations are limited to reports from 0 to 1.6 nmi.
PTDY	Pressure Tendency is the direction (plus or minus) and the amount of pressure change (hPa)for a three hour period ending at the time of observation. (not in Historical files)
TIDE	The water level in feet above or below Mean Lower Low Water (MLLW) .

Standard Meteorological Data for Drifting Buoys

#YY	MM	DD	hhmm	LAT	LON	WDIR	WSPD	GST	PRES	PTDY	ATMP	WTMP	DEWP	I
#yr	mo	dy	hrmn	deg	deg	degT	m/s	m/s	hPa	hPa	degC	degC	degC	
2022	07	27	1200	25.99	-89.66	60	9.0	MM	1010.1	MM	25.2	24.5	24.2	



This format is also used for uncrewed surface vehicles.

LAT	Latitude in decimal degrees at which observation was taken. Positive values represent northern hemisphere while negative values represent southern hemisphere.
LON	Longitude in decimal degrees at which observation was taken. Positive values represent eastern hemisphere while negative values represent western hemisphere.
WDIR	Wind direction (the direction the wind is coming from in degrees clockwise from true N) during the same period used for WSPD. See Wind Averaging Methods
WSPD	Wind speed (m/s) averaged over an eight-minute period for buoys and a two-minute period for land stations. Reported Hourly. See Wind Averaging Methods .
GST	Peak 5 or 8 second gust speed (m/s) measured during the eight-minute or two-minute period. The 5 or 8 second period can be determined by payload, See the Sensor Reporting, Sampling, and Accuracy section.
PRES	Sea level pressure (hPa). For C-MAN sites and Great Lakes buoys, the recorded pressure is reduced to sea level using the method described in <i>NWS Technical Procedures Bulletin 291</i> (11/14/80). (labeled BAR in Historical files)
PTDY	Pressure Tendency is the direction (plus or minus) and the amount of pressure change (hPa) for a three hour period ending at the time of observation. (not in Historical files)
ATMP	Air temperature (Celsius). For sensor heights on buoys, see Hull Descriptions . For sensor heights at C-MAN stations, see C-MAN Sensor Locations
WTMP	Sea surface temperature (Celsius). For buoys the depth is referenced to the hull's waterline. For fixed platforms it varies with tide, but is referenced to, or near Mean Lower Low Water (MLLW) .
DEWP	Dew point temperature taken at the same height as the air temperature measurement.
WVHT	Significant wave height (meters) is calculated as the average of the highest one-third of all of the wave heights during the 20-minute sampling period. See the Wave Measurements section.
DPD	Dominant wave period (seconds) is the period with the maximum wave energy. See the Wave Measurements section.

Derived Met Values

#YY	MM	DD	hh	mm	CHILL	HEAT	ICE	WSPD10	WSPD20
#yr	mo	dy	hr	mn	degC	degC	cm/hr	m/s	m/s
2014	09	11	16	50	MM	34.4	MM	5	5

HEAT For more information on heat index, please see the [NWS Heat Wave](#) page.

WSPD	Ten-minute average wind speed values in m/s. (SPD in Historical files)
GDR	Direction, in degrees clockwise from true North, of the GST, reported at the last hourly 10-minute segment.
GST	Maximum 5-second peak gust during the measurement hour, reported at the last hourly 10-minute segment.
GTIME	The minute of the hour that the GSP occurred, reported at the last hourly 10-minute segment.

For more information on continuous winds and the timing of these measurements, see the [continuous winds](#) help section.

Detailed Wave Summary (Realtime data files only)

#YY	MM	DD	hh	mm	WVHT	SwH	SwP	WWH	WWP	SwD	WWD	STEEPNESS	APD	MWD
#yr	mo	dy	hr	mn	m	m	sec	m	sec	-	degT	-	sec	degT
2014	09	11	17	00	0.6	0.4	5.6	0.4	4.3	SE	MM	N/A	4.2	134

WVHT	Significant Wave Height is the average height (meters) of the highest one-third of the waves during a 20 minute sampling period.
SwH	Swell height is the vertical distance (meters) between any swell crest and the succeeding swell wave trough.
SwP	Swell Period is the time (usually measured in seconds) that it takes successive swell wave crests or troughs pass a fixed point.
WWH	Wind Wave Height is the vertical distance (meters) between any wind wave crest and the succeeding wind wave trough (independent of swell waves).
WWP	Wind Wave Period is the time (in seconds) that it takes successive wind wave crests or troughs to pass a fixed point.
SwD	The direction from which the swell waves at the swell wave period (SWPD) are coming. The units are degrees from true North, increasing clockwise, with North as 0 (zero) degrees and East as 90 degrees.
WWD	The direction from which the wind waves at the wind wave period (WWPD) are coming. The units are degrees from true North, increasing clockwise, with North as 0 (zero) degrees and East as 90 degrees.
STEEPNESS	Wave steepness is the ratio of wave height to wave length and is an indicator of wave stability. When wave steepness exceeds a 1/7 ratio; the wave becomes unstable and begins to break.
APD	Average Wave Period is the average period (seconds) of the highest one-third of the wave observed during a 20 minute sampling period.
MWD	The direction from which the waves at the dominant period (DPD) are coming. The units are degrees from true North, increasing clockwise, with North as 0 (zero) degrees and East as 90 degrees. See the Wave Measurements section.

Spectral Wave Data

```
#YY MM DD hh mm Sep_Freq < spec_1 (freq_1) spec_2 (freq_2) spec_3 (freq_3)
2014 09 11 17 00 0.225 0.000 (0.033) 0.000 (0.038) 0.000 (0.043) ...>
```



```
#YY MM DD hh mm alpha1_1 (freq_1) alpha1_2 (freq_2) alpha1_3 (freq_3) ... >
2014 09 11 17 00 999.0 (0.033) 999.0 (0.038) 999.0 (0.043) ...>
```



```
#YY MM DD hh mm alpha2_1 (freq_1) alpha2_2 (freq_2) alpha2_3 (freq_3) ... >
2014 09 11 17 00 999.0 (0.033) 999.0 (0.038) 999.0 (0.043) ...
```



```
#YY MM DD hh mm r1_1 (freq_1) r1_2 (freq_2) r1_3 (freq_3) ... >
2014 09 11 17 00 999.00 (0.033) 999.00 (0.038) 999.00 (0.043) ...>
```

```
#YY MM DD hh mm r2_1 (freq_1) r2_2 (freq_2) r2_3 (freq_3) ... >
2014 09 11 17 00 999.00 (0.033) 999.00 (0.038) 999.00 (0.043) ...>
```

Sep_Freq The Separation Frequency is the frequency that separates wind waves (WWH, WWP, WWD) from swell waves (SWH, SWP, SWD). NDBC inserts the value 9.999 if Sep_Freq is missing.

Spectral wave density Energy in (meter*meter)/Hz, for each frequency bin (typically from 0.03 Hz to 0.40 Hz).

Spectral wave direction Mean wave direction, in degrees from true North, for each frequency bin. A list of [directional stations](#) is available.

Directional Wave Spectrum = $C11(f) * D(f,A)$, f =frequency (Hz), A =Azimuth angle measured clockwise from true North to the direction wave is from.
 $D(f,A) = (1/\pi) * (0.5 + R1 * \cos(A - \text{ALPHA1}) + R2 * \cos(2 * (A - \text{ALPHA2})))$. $R1$ and $R2$ are the first and second normalized polar coordinates of the Fourier coefficients and are nondimensional. ALPHA1 and ALPHA2 are respectively mean and principal wave directions.

In terms of Longuet-Higgins Fourier Coefficients

- $R1 = (\text{SQRT}(a_1^2 + b_1^2)) / a_0$
- $R2 = (\text{SQRT}(a_2^2 + b_2^2)) / a_0$
- $\text{ALPHA1} = 270.0 - \text{ARCTAN}(b_1, a_1)$
- $\text{ALPHA2} = 270.0 - (0.5 * \text{ARCTAN}(b_2, a_2) + \{0. \text{ or } 180.\})$

Notes:

1. The *R1* and *R2* values in the monthly and yearly historical data files are scaled by 100, a carryover from how the data are transported to the archive centers. The units are hundredths, so the *R1* and *R2* values in those files should be multiplied by 0.01.
2. *D(f,A)* can take on negative values because of the trigonometric sine and cosine functions. There are several approaches to prevent or deal with the negative values. For more information and discussion of some approaches see: Use of advanced directional wave spectra analysis methods, M. D. Earle, K. E. Steele, and D. W. C. Wang, Ocean Engineering, Volume 26, Issue 12, December 1999, Pages 1421-1434.
3. *ALPHA2* has ambiguous results in using the arctangent function with the Fourier Coefficients, b_2 , a_2 . When necessary, NDBC adds 180 degrees to *ALPHA2* in order to minimize the difference between *ALPHA1* and *ALPHA2*.

For more information on the mathematics behind the measuring of surface water waves, see the [waves](#) help section.

Ocean Current Data

```
#YY  MM DD hh mm DEP01 DIR01 SPD01 DEP02 DIR02 SPD02 DEP03 DIR03 SPD03 ...>
#yr  mo dy hr mn      m  degT  cm/s      m  degT  cm/s      m  degT  cm/s ...>
2014 09 11 17 04      2   40    8      10   120    5      14   250   13 ...>
```

DEP01, DEP02, ... The distance from the sea surface to the middle of the depth cells, or bins, measured in meters.

DIR01, DIR02, ... The direction the ocean current is flowing toward. 0-360 degrees, 360 is due north, 0 means no measurable current.

SPD01, SPD02, ... The speed of the ocean current measured in cm/s.

Ocean Current Data (Expanded ADCP format)

```
#YY  MM DD hh mm I Bin  Depth Dir Speed ErrVl VerVl %Good3 %Good4 %GoodE
#yr  mo dy hr mn -  -    m  degT  cm/s  cm/s  cm/s      %      %      %
2014 09 11 17 46 1   1   69.4 117  63.2  -0.7  -1.2    0    100    0
2014 09 11 17 46 1   2  101.4 122  63.1  -1.0  -3.7    0    100    0
2014 09 11 17 46 1   3  133.4 120  54.1   4.2  -3.4    0    100    0
```



Instrument Number Stations may have more than one ADCP instrument. This field distinguishes these instruments by number. Valid values are 0-9, with 0 being reserved for surface measurements.

Bin The bin number, ranging from 1 to 128, where 1 is the bin closest to the transducer head.

Depth	The distance from the sea surface to the middle of the depth cells, or bins, measured in meters.
Dir	The direction the ocean current is flowing toward. 0-360 degrees, 360 is due north, 0 means no measurable current.
Speed	The speed of the ocean current measured in cm/s.
ErrVl	The error velocity measured in cm/s.
VerVl	The vertical velocity of the ocean current measured in cm/s.
%Good3	The percentage of three-beam solutions that are good.
%Good4	The percentage of four-beam solutions that are good.
%GoodE	The percentage of transformations rejected.
EI1, EI2, EI3, EI4	The echo intensity values for the four beams. Valid values are 0 to 255. EI1 = Echo Intensity for beam #1; EI2 = Echo Intensity for beam #1; EI3 = Echo Intensity for beam #3; and EI4 = Echo Intensity for beam #4.
CM1, CM2, CM3, CM4	The correlation magnitude values for the four beams. Valid values are 0 to 255. CM1 = Correlation Magnitude for beam #1; CM2 = Correlation Magnitude for beam #1; CM3 = Correlation Magnitude for beam #3; and CM4 = Correlation Magnitude for beam #4.
Flags	The nine quality flags represent the results of the following quality tests based on their position in the flags field. Flag 1 represents the overall bin status. Flag 2 represents the ADCP Built-In Test (BIT) status. Flag 3 represents the Error Velocity test status. Flag 4 represents the Percent Good test status. Flag 5 represents the Correlation Magnitude test status. Flag 6 represents the Vertical Velocity test status. Flag 7 represents the North Horizontal Velocity test status. Flag 8 represents the East Horizontal Velocity test status. Flag 9 represents the Echo Intensity test status. Valid values are: 0 = quality not evaluated; 1 = failed quality test; 2 = questionable or suspect data; 3 = good data/passed quality test; and 9 = missing data.

Marsh-McBirney Current Measurements

YY	MM	DD	hh	mm	DIR	SPD
96	10	31	23	0	198	1.1

DIR Direction the current is flowing TOWARDS, measured in degrees clockwise from North.

SPD Current speed in cm/s.

Water Level

#YY	MM	DD	hh	mm	TG01	TG02	TG03	TG04	TG05	TG06	TG07	TG08	TG09	TG10
2014	07	01	00	00	10.6	10.6	10.6	10.5	10.6	10.6	10.6	10.7	10.7	10.8

TG01, TG02, ..., TG10 Six-minute water levels representing the height, in feet, of the water above or below [Mean Lower Low Water \(MLLW\)](#), offset by 10 ft. to prevent negative values. Please subtract 10 ft. from every value to obtain the true water level value, in reference to MLLW.

Oceanographic Data

#YY	MM	DD	hh	mm	DEPTH	OTMP	COND	SAL	O2%	O2PPM	CLCON	TURB	PH
#yr	mo	dy	hr	mn	m	degC	mS/cm	psu	%	ppm	ug/l	FTU	-
2014	09	11	17	00	1.0	29.05	MM	34.98	MM	MM	MM	MM	MM



Depth (DEPTH) Depth (meters) at which measurements are taken.

Ocean Temperature (OTMP) The direct measurement (Celsius) of the Ocean Temperature (as opposed to the indirect measurement (see WTMP above)).

Conductivity (COND) Conductivity is a measure of the electrical conductivity properties of seawater in milliSiemens per centimeter.

Salinity (SAL) Salinity is computed by a known functional relationship between the measured electrical conductivity of seawater (CON), temperature (OTMP) and pressure. Salinity is computed using the Practical Salinity Scale of 1978 (PSS78) and reported in Practical Salinity Units.

Oxygen Saturation (O2%) Dissolved oxygen as a percentage.

Oxygen Concentration (O2PPM) Dissolved oxygen in parts per million.

Chlorophyll Concentration (CLCON) Chlorophyll concentration in micrograms per liter (ug/l).

Turbidity (TURB)	Turbidity is an expression of the optical property that causes light to be scattered and absorbed rather than transmitted in straight lines through the sample (APHA 1980). Units are Formazin Turbidity Units (FTU).
pH (PH)	A measure of the acidity or alkalinity of the seawater.
Eh (EH)	Redox (oxidation and reduction) potential of seawater in millivolts.

Solar Radiation Data

```
#YY  MM DD hh mm  SRAD1  SWRAD  LWRAD
#yr  mo dy hr mn   w/m2   w/m2   w/m2
2014 09 11 18 00 1061.0    MM    MM
```

Shortwave Radiation (SRAD1, SWRAD)	Average shortwave radiation in watts per square meter for the preceding hour. Sample frequency is 2 times per second (2 Hz). If present, SRAD1 is from a LI-COR LI-200 pyranometer sensor, and SWRAD is from an Eppley PSP Precision Spectral Pyranometer.
Longwave Radiation (LWRAD)	Average downwelling longwave radiation in watts per square meter for the preceding hour. Sample frequency is 2 times per second (2 Hz). If present, LWRAD is from an Eppley PIR Precision Infrared Radiometer.

DART® (Tsunameters) Measurements

```
#YY  MM DD hh mm ss T  HEIGHT
#yr  mo dy hr mn s  -   m
2014 09 11 17 00 00 1 5848.422
```

T (TYPE)	Measurement Type: • 1 = 15-minute measurement; • 2 = 1-minute measurement; and • 3 = 15-second measurement.
----------	---

HEIGHT	Height of water column in meters.
--------	-----------------------------------

tt = Tsunami Trigger Time, see the [Tsunami Detection Algorithm](#)

ts = data Time Stamp(s)

24-Hour Rain Measurements

```
#YY  MM DD hh mm  RATE  PCT  SDEV
#yr  mo dy hr mn   mm/h   %    -
2008 01 01 12 00   0.0   0.0   0.1
```

24-Hour Rain Rate	Average precipitation rate in units of millimeters per hour over 24-hour period from 00:00 to 23:59.99 GMT.
-------------------	---

Percent Time Raining in 24-Hour Period	Percentage of 144 ten-minute periods within a 24 hour period with a measurable accumulation of precipitation.
SDev	---
Flag	In the case of 24-hour rainfall measurements, a flag is assigned when over half of the 10-minute measurements from which it is derived are flagged.

Hourly Rain Measurements

```
#YY MM DD hh mm ACCUM
#yr mo dy hr mn mm
2008 01 01 00 30 0.0
```

Hourly Rain Accumulation	Total accumulation of precipitation in units of millimeters on station during the 60-minute period from minute 0 to minute 59:59.99 of the hour.
Flag	In the case of one-hour accumulation, a flag is assigned when over half of the 10-minute measurements from which it is derived have been flagged.

10-Minute Rain Measurements

```
#YY MM DD hh mm RATE
#yr mo dy hr mn mm/h
2008 01 01 00 00 0.0
```

10-Minute Rain Rate	Rain rate in units of millimeters per hour on station over the 10-minute period from 5 minutes before to 4 minutes 59.99 seconds after the time with which it is associated.
Flag	In the case of 10-minute rainfall measurements, a flag is assigned to any measurement when either the -5 or +5 minute rain measurement from which it is derived is missing or obviously an error.

Housekeeping Measurements

```
#YY MM DD hh mm BATTV BATTCURR BATTTEMP REMCAP
#yr mo dy hr mn Volts Amps DegC Ah
2016 09 15 19 00 12.381 -0.177 32.9 116.8
```

BATTV	Hourly Average Battery Voltage (volts)
BATTCURR	Hourly Average Battery Current (amperes)
BATTTEMP	Hourly Average Battery Temperature (degrees Celsius)
REMCAP	Remaining Battery Capacity (ampere-hours)

Discontinued Measurement Abbreviations

Some historical files have column heading abbreviations that have changed over time. The old abbreviations are listed below with links to the new standardized abbreviation description.

Old	New Abbreviation
WD	WDIR - Wind Direction
DIR	WDIR - 10 Minute Wind Direction
SPD	WSPD - 10 Minute Wind Speed
GSP	GST - Gust in Continuous Winds data
GMN	GTIME - Time of Gust in Continuous Winds data
BARO	PRES - Pressure
H0	WVHT - Significant Wave Height
DOMPD	DPD - Dominant Wave Period
AVP	APD - Average Wave Period
SRAD	SWRAD - Short Wave Solar Radiation
SRAD2	SWRAD - LI-COR Short Wave Solar Radiation
LRAD	LWRAD - Long Wave Solar Radiation
LRAD1	LWRAD - Long Wave Solar Radiation

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